Yuxing Yang

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Education

University of Illinois Urbana-Champaign Aug 2024 – May 2027
B.S. Information Sciences + Data Science GPA: 3.97/4.0 *Champaign, Illinois*
Edmund J. James Scholar; Dean's List; **ITI Cybersecurity Certificate**.

Research Interests

Post-quantum cryptography and crypto-agility; security measurement and usable deployment; cybersecurity policy and governance; AI governance and security; privacy by design; children's privacy and age-appropriate design; Contextual Integrity; AI-generated text provenance and digital libraries.

Academic Profile

Interdisciplinary researcher working across post-quantum cryptography, cybersecurity systems, AI governance, privacy, and information policy.
Research record: first- and second-place 2026 ITC Student Paper Awards, a 2025 ITC award, an ASIS&T long paper, an ACM/IEEE JCDL poster, and accepted work in privacy, smart-grid security, and cyber-physical systems.

Awards & Honors

Cliff Aggen Undergraduate Student Paper 1st Place Award	ITC 2026
Cliff Aggen Undergraduate Student Paper 2nd Place Award	ITC 2026
Cliff Aggen Student Paper Award, Second Place	ITC 2025
G Edwards Evans Living Trust Scholarship	2026–2027
Edmund J. James Scholar; Dean's List	UIUC

Publications & Scholarly Presentations

Peer-Reviewed Conference Papers and Proceedings

[1] **Y. Yang**, "Implementing and Evaluating Post-Quantum Key Establishment in Telemetry Systems," *International Telemetering Conference (ITC)*, 2026. **Accepted; Cliff Aggen Undergraduate Student Paper 1st Place Award.**

[2] **Y. Yang**, S. Cui, and K. Wei, "QuantumSense: An Adaptive ML-KEM Scheme Selection Framework for Latency-Constrained Satellite Telemetry Pipelines," *International Telemetering Conference (ITC)*, 2026. **Accepted.**

[3] T. Randall, J. Wilcken, and **Y. Yang**, "Radio Implementation and Interception of a McEliece-Based Cryptosystem," *International Telemetering Conference (ITC)*, 2026. **Accepted; Cliff Aggen Undergraduate Student Paper 2nd Place Award.**

[4] **Y. Yang**, "HQC as a Backup KEM for Operational Technology: Algorithm Diversity in Long-Lifecycle Industrial Systems," *EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP)*, 2026. **Accepted.**

[5] **Y. Yang**, "A Lightweight Federated Edge Intelligence Framework for Anomaly Detection in Smart Grid IoT Networks," *EAI International Conference on Smart Grid and Innovative Frontiers in Telecommunications (SmartGIFT)*, 2026. **Accepted for presentation and publication.**

[6] **Y. Yang**, "Stylometric Analysis of AI-Generated Text: Lexical Fidelity vs. Structural Limitations," *Association for Information Science and Technology (ASIS&T)*, 2026. **Accepted long paper.**

[7] **Y. Yang**, "Age-Sensitive Privacy Expectations on Social Media: A Contextual Integrity Analysis," *International Conference on Social Media Technologies, Communication, and Informatics (SOTICS)*, 2026. **Accepted short paper / work in progress.**

[8] **Y. Yang**, "Recovering the Source of AI-Transformed Text: Retrieval and Residual Traces in a Digital Collection," *ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, 2026. **Accepted poster paper.**

[9] B. Arnesen, T. Wai, T. Randall, **Y. Yang**, D. G. M. Mitchell, and W. K. Harrison, "Applying Post-Quantum Cryptography Based on Quasi-Cyclic Codes to Aeronautical Mobile Telemetry," *International Telemetering Conference Proceedings*, vol. 60, 2025. `[paper]`

Invited and Non-Archival Scholarly Presentations

[1] **Y. Yang**, "Contextual Integrity and Age: Rethinking Transmission Principles Across Age Groups," *8th Annual PrivaCI Symposium*, 2026. Accepted for a dedicated Contextual Integrity Community Feedback Session.

[2] **Y. Yang** and A. Nikolich, "Quantifying AI/ML Adoption in Science: Mapping Scientific Workflows, Models, and Autonomy," *UChicago and Caltech AI + Science Conference*, Chicago, IL, October 19–21, 2026. **Accepted**.

Research Experience

Cryptanalysis of Concatenated McEliece Cryptosystems Aug 2025 – Present
Researcher, ICE Lab, BYU · Advisors: W. K. Harrison & D. G. M. Mitchell *Provo, Utah / Remote*

- Develop a **two-stage cryptanalytic framework** combining Information Set Decoding with an Attack on the Sum of Two Codewords against McEliece variants constructed from depth-two concatenated binary codes.
- Analyze parameter sets proposed by Branco et al., Wang et al., and Lin et al.; evaluate whether LDPC or QC-LDPC inner-code structure reduces effective security below claimed levels.
- Extend the attack analysis across different outer-code constructions, including Goppa- and Polar-based designs, and derive work-factor expressions and bounds associated with relevant null-space dimensions.
- Design parallel, block-wise attack strategies that exploit quasi-cyclic structure and compare their complexity with conventional dual-code and information-set-decoding attacks.
- Run reproducible simulations across published parameter sets to compare theoretical and empirical work factors and identify implementation-sensitive security assumptions.

Post-Quantum Cryptography for Aeronautical and Satellite Telemetry May 2025 – Present
Researcher, ICE Lab, BYU · Advisors: W. K. Harrison & D. G. M. Mitchell *Provo, Utah / Remote*

- Investigate deployment of **BIKE, HQC, and ML-KEM** for secure key establishment in aeronautical mobile telemetry and satellite communication environments.
- Analyze public-key and ciphertext sizes, decoding-failure considerations, computational overhead, and resistance to information-set-decoding and reaction-style attacks.
- Map code-based key-encapsulation mechanisms to the operational constraints of the iNET architecture and long-lifecycle telemetry systems.
- Contributed to a 2025 ITC proceedings paper and subsequent 2026 work on adaptive ML-KEM selection and radio implementation/interception of McEliece-based cryptosystems.
- Selected for the competitive **BYU Immerse-X Undergraduate Research Program**; related papers received ITC student-paper recognition in 2025 and 2026.

Post-Quantum Key Establishment Testbed Jan 2026 – May 2026
Edmund J. James Scholar Project, UIUC ITI · Faculty Advisor: Dominic Saebeler *Champaign, Illinois*

- Independently designed and executed a study of post-quantum migration as an operational security and infrastructure problem.
- Built a software testbed for Ground Station–Test Article key establishment using **TLS 1.3, OpenSSL, and the Open Quantum Safe provider**; evaluated ECDH, three ML-KEM security levels, and hybrid ECDH/ML-KEM configurations.
- Measured mean and tail latency, CPU time, memory use, and communication overhead across controlled round-trip time, bandwidth, and concurrent-load conditions.
- Identified implementation bias created by comparing algorithms through unmatched software libraries and redesigned the testbed so every configuration ran through the same TLS stack.
- Found that ECDH and ML-KEM-512 differed by less than **2%** under matched baseline conditions, while bandwidth constraints and concurrent hybrid handshakes produced more consequential deployment costs.
- Authored the resulting sole-author paper, which was accepted to **ITC 2026** and received the **Cliff Aggen Undergraduate Student Paper 1st Place Award**.

Children's Privacy Harm and the UK Age Appropriate Design Code Sep 2026 – Present
Research Project, School of Information Sciences, UIUC *Champaign, Illinois*

- Conduct deductive qualitative coding of the UK **Age Appropriate Design Code's 15 standards** to identify forms of privacy harm anticipated by child-centered data-protection regulation.
- Map concepts including best interests, privacy by default, data minimization, profiling, geolocation, transparency, parental controls, and nudge techniques to platform design and data practices.
- Examine the relationship between legal duties, observable interface behavior, internal technical evidence, and the limits of external compliance auditing.
- Develop structured coding materials and a traceability framework connecting protected interests, anticipated harms, regulated practices, evidentiary requirements, and residual legal judgment.

Age-Sensitive Contextual Integrity and Social-Media Privacy Aug 2025 – Present
Researcher, School of Information Sciences, UIUC · Advisor: Madelyn R. Sanfilippo *Champaign, Illinois*

- Developed an age-sensitive extension of *Contextual Integrity*, treating age as a structuring condition that changes the normative meaning of information flows.
- Introduced **normative homogeneity** as a limitation of privacy frameworks that collapse age-differentiated expectations into an adult-centered default.
- Modeled how actors, attributes, transmission principles, and privacy expectations vary across adolescent and adult social-media contexts.
- Presented early work at the **8th Annual PrivaCI Symposium**; a related short paper was accepted to **SOTICS 2026**.

AI-Generated Text Stylometry and Source Recovery Jan 2026 – Present
Researcher, School of Information Sciences, UIUC · Mentor: Edwin (Teddy) Roland *Champaign, Illinois*

- Developed a matching-versus-nonmatching distance framework for evaluating AI stylistic imitation beyond binary human/AI classification.
- Conducted controlled experiments across ChatGPT, Claude Sonnet, and Gemini Pro, measuring lexical, syntactic, sentence-level, and structural features in human and AI-transformed text.
- Identified a statistically significant **lexical–structural asymmetry**: models reproduced word-level characteristics more faithfully than sentence-length distributions.
- Extended the work to retrieval and source attribution, studying whether AI transformation preserves residual signals sufficient to recover an original item from a digital collection.
- Produced an **ASIS&T 2026 long paper** and an **ACM/IEEE JCDL 2026 poster paper**; presented work through the iSchool Student Research Showcase.

AI/ML Adoption across Scientific Disciplines Aug 2026 – Present
Independent Study, School of Information Sciences, UIUC · Advisor: Anita Nikolich *Champaign, Illinois*

- Build a taxonomy of AI use across scientific workflow stages, capability levels, and disciplinary settings.
- Distinguish applications ranging from traditional machine learning and scientific foundation models to autonomous agents, AI scientists, robotic laboratories, and closed-loop experimentation systems.
- Develop a structured dataset using OpenAlex, Semantic Scholar, public funding databases, research-center materials, arXiv/bioRxiv literature, and other reproducible sources.
- Design a Python/Jupyter workflow for API-based collection, deduplication, metadata enrichment, screening, and human-validated coding of relevant publications and projects.
- Contribute foundational evidence for broader research on security threats and governance requirements in AI-enabled scientific workflows.

EACUE Community Web Application Jan 2026 – Present
Software Developer, NCSA, UIUC · Supervisor: Matthew John Berry *Champaign, Illinois*

- Contribute to a community-facing application supporting structured data collection and public-engagement initiatives.
- Configured and debugged a full-stack environment using **Next.js, TypeScript, Bun, MongoDB, and Docker**.

- Traced backend and runtime failures involving database connectivity, environment variables, application state, and missing seed data.
- Analyze authentication, API, and survey-data workflows and collaborate with stakeholders on question structure, content, and interface revisions.

Asia-Pacific Youth Debrief for COP29 Jul 2024 – Dec 2024
Developer and Youth Delegate, UNDP Asia-Pacific / YECAP Collaboration *Beijing, China / Remote*

- Co-developed a digital platform used to synthesize and communicate youth perspectives on climate and development priorities.
- Collaborated with more than **300 participants** and partner organizations and represented project findings in connection with COP29 youth-engagement activities.

Professional Experience

University Housing, University of Illinois Urbana-Champaign Aug 2026 – Present
Resident Advisor *Champaign, Illinois*

- Support residents in a university residence hall through community programming, policy education, individual advising, and referral to campus resources.
- Respond to time-sensitive safety, wellness, and interpersonal incidents; document events accurately and coordinate escalation with professional housing staff and emergency personnel.
- Plan and facilitate floor meetings, communicate operational updates across in-person and online channels, and promote an inclusive residential environment.

Learningly AI Aug 2025 – Present
Chief Operating Officer *Evanston, Illinois*

- Lead operations and strategic development for an AI-enabled personalized-learning platform, coordinating product decisions, partnerships, execution priorities, and team workflows.
- Address product-governance questions involving student data protection, algorithmic transparency, and responsible use of AI in educational settings.
- Venture selected for the **USC EdTech Accelerator, Cohort 8**.

Ourfreedom.ai Oct 2025 – Present
Software and Network Security Engineer Intern *Remote / New York*

- Develop secure application features supporting communication between incarcerated individuals and family members using **React Native and Firebase**.
- Contribute to authentication, data-protection, network-security, and privacy-sensitive design decisions in a high-risk operational context.
- Balance security controls with usability requirements for users facing constrained devices, access, and communication environments.

China Mobile International Limited Jul 2025 – Aug 2025
IT Project Manager Intern *Hybrid / China*

- Coordinated cross-functional work supporting deployment and evaluation of 5G communication products.
- Built performance dashboards and analyzed adoption, traffic, and system KPI data to inform product and operational decisions.
- Assessed market outcomes during the internship period; Jiangsu Mobile's 5G user penetration reached **76%**, a four-percentage-point increase over the pre-summer baseline.

Shanghai Enlead Technology Co., Ltd. Sep 2024 – Present
Data Analyst Intern and Head Judge *Shanghai, China / Remote*

- Analyze participation, performance, and market data to support strategic planning for debate education and tournament operations.
- Serve as Head Judge for more than **20 public-forum debate tournaments**, supervising adjudication standards and providing structured feedback.

- Supported organizational collaboration with NSDA U.S. and the expansion of WSDA's national and international programming.

ATLAS Internship Program, UIUC Spring 2026
Client Relations Specialist, CRM Team *Champaign, Illinois*

- Researched prospective community clients, documented interactions, and developed communications explaining student-intern capabilities and project-matching opportunities.
- Participated in Champaign County Community Coalition meetings and supported LAS Public Engagement initiatives, including the Educating Character Initiative and the Alma Project.

Teaching Experience

CS 124: Introduction to Computer Science I, UIUC Aug 2025 – Dec 2025
Tutor and Course Assistant *Champaign, Illinois*

- Tutored students in Java programming, object-oriented design, debugging, decomposition, and algorithmic problem solving.
- Improved course web materials using HTML and human–computer interaction principles to strengthen clarity, usability, and accessibility.

Leadership & Service

Disability Education Media Project, UIUC Jan 2026 – Present
Media Production Assistant *Champaign, Illinois*

- Supervisor: **Emily Tarconish**. Support production and editing of interview-based educational media featuring students with disabilities.
- Improve visual framing, subject positioning, narrative clarity, and accessibility for community-engaged digital storytelling.

Champaign YSA Ward Feb 2025 – Present
Elders Quorum Presidency; Sunday School Teacher *Champaign, Illinois*

- Serve in congregational leadership for The Church of Jesus Christ of Latter-day Saints and teach recurring Sunday lessons for a young-adult faith community of more than 80 members.
- Support member outreach, service coordination, discussion-based learning, and community care.

Selected Professional Development

Goldman Sachs Summer Possibilities Summits Dec 2024 – May 2025
Participant and Team Leader *New York, New York*

- Selected for programming focused on professional development, leadership, financial services, and engagement with industry practitioners.

Goldman Sachs Virtual Insight Series May 2025
Participant *New York, New York*

- Completed a four-week program examining Goldman Sachs' business divisions, organizational culture, and career pathways.

Goldman Sachs Asia Possibilities Series: Finance Applications of AI May 2025
Participant *New York, New York*

- Participated in workshops on AI applications, financial innovation, and emerging technology across business functions.

Goldman Sachs Asset Management Insight Day May 2025
Participant *New York, New York*

- Participated in seminars on asset-management strategies, investment research, market trends, and career pathways.

Point72 Academy Spring Sessions Apr 2025 – May 2025
Participant *Stamford, Connecticut*

– Selected for seminars on hedge-fund fundamentals, investment research, analytical decision making, and career development.

PeakSpan Capital Growth Equity Immersion Program Dec 2025 – Jan 2026
Participant *New York, New York*

– Completed a four-part program covering growth-stage investing, financial modeling, market analysis, and venture evaluation.

IYC 11: Youth Shaping Global Development Apr 2025 – May 2025
Participant *Los Angeles, California*

– Participated in an international youth program focused on sustainable development, diplomacy, and impact-oriented innovation.

Relevant Coursework

Completed — Cybersecurity and Systems: Introduction to Risk and Cybersecurity; Introduction to Network Systems.

Completed — Computer Science and Data Science: Introduction to Computer Science I (Java); Data Science Discovery; Data Science Exploration; AI Systems in Games; Global FinTech Analytics: Ethics.

Completed — Research and Information Science: Introduction to Information Sciences; Social Aspects of Information Technology; Analytical Foundations of Information Problems; Research Design in Information Sciences; Introduction to Database Concepts and Applications; Computing in the Humanities; Data Storytelling.

Completed — Mathematics, Statistics, and Social Science: Calculus I; Calculus II; Linear Algebra; Mathematical Statistics; Microeconomic Principles; Macroeconomic Principles.

In Progress (Fall 2026): Algorithms and Data Structures for Data Science; Concepts of Machine Learning; Ethics and Policy for Data Science; Data Management, Curation, and Reproducibility; Reading and Writing Data; Global Development and Environment.

Planned: Discrete Structures; Foundations of Information Security; additional upper-level statistics coursework.

Technical & Research Skills

Programming and Analysis: Python, Java, R, TypeScript, JavaScript, SQL, HTML; Jupyter Notebook; statistical hypothesis testing; machine-learning evaluation; qualitative and deductive coding.

Cryptography and Security: McEliece, BIKE, HQC, ML-KEM, hybrid key establishment, TLS 1.3, Information Set Decoding, Dual Code Attack, LDPC/QC-MDPC codes, security work-factor analysis, crypto-agility, and migration assessment.

Data and Research Infrastructure: OpenAlex and Semantic Scholar APIs; reproducible data collection; metadata normalization; experiment design; latency and resource measurement; stylometric feature extraction; retrieval and source attribution.

Web and Software: Next.js, React Native, Firebase, MongoDB, Bun, Docker, REST/API workflows, authentication, and survey-data pipelines.

Tools: Git/GitHub, LaTeX, GNU Radio, OpenSSL, Open Quantum Safe, Microsoft Office, Microsoft Excel.

Policy and Governance Methods: Contextual Integrity; privacy-harm analysis; regulatory coding; privacy-by-design assessment; cybersecurity risk analysis; technical-policy translation.

Additional Interests

Creative writing; archery; guitar; public-forum debate; international trade and technology policy.